

Name: Priyangan Vasantharajan

Student ID: 25197298

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Note: Lab_1_2 Exercise

Question : 3.2 (Lab 1 Submission Exercises)

The Pseudocode

1. Start
2. Use variables: A, B, C root1 and root2 as float
3. Input values for A, B and C
4. Print the input values for A, B and C
5. Calculate the discriminant $d = \sqrt{B^2 - 4AC}$
6. Calculate the roots
 - $\text{root1} = (-B + d) / (2 * A)$
 - $\text{root2} = (-B - d) / (2 * A)$
7. Print root1, root2
8. End

Tested Given Questions with Answers
(according to the spyder app)

A = 1, B = 0, C = -4

Root #1 = 2.0

Root #2 = -2.0

A = 1, B = 5, C = -36

Root #1 = 4

Root #2 = -9

A = 2, B = 7.5, C = 6

Root #1 = -1.15

Root #2 = -2.59

A = 0, B = 3.5, C = 8 # this test case fails and generates an error (why?):

Root #1 = error

Root #2 = error

because: python gives error message for $d.\text{sqrt}()$ negative number.

solution: we need add discriminant check.